

Linear Regression 3 : Pre-reads

1. Importance of Train-Test Split

Description: Split dataset into training (to fit the model) and testing (to evaluate performance).

Importance:

- Prevents overfitting.
 - Provides an unbiased estimate of how the model generalizes to unseen data.
-

2. Feature Importance and Interpretation

Description: Measures how much each feature contributes to the prediction.

Importance:

- Improves interpretability.
 - Helps identify key drivers (e.g., “engine size” → mileage).
-

3. Importance of Standardization

Description: Scaling features to have mean = 0, standard deviation = 1.

Importance:

- Essential for algorithms that depend on distance/gradient (e.g., regression, SVM, neural nets).
 - Prevents dominance of large-scale features.
-

4. Min-Max Scaler

Description: Scales values between 0 and 1, in simple words, it Normalizes the values.

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Importance: Useful when features have different units but bounded scales are needed.

5. Case Study: Predict Mileage of a Car

Goal: Predict “miles per gallon (MPG)” using features like horsepower, weight, cylinders, etc.

Importance: Hands-on example of regression in action, combining preprocessing, encoding, scaling, and model training.

6. Loading Data

Description: Import datasets via CSV/SQL/Excel.

Importance: First step in any ML workflow.

7. One-Hot Encoding

Description: Represent categorical variables as binary vectors.

Importance: Required for ML models (like regression) that need numeric input.

8. Create Dummy Variables

Description: Alternative to one-hot, creates binary columns for categories.

Importance: Handles categorical data in regression.

9. Handling Missing Values

Approaches:

- Drop rows/columns.
- Impute with mean/median/mode.
- Use domain-specific strategies.
Importance: Missing data can bias results and reduce accuracy.

10. Data Splitting

Description: Split into train/validation/test sets.

Importance: Provides checkpoints to evaluate performance before deployment.

11. Fit Linear Model

Description: Train regression model on prepared data.

Importance: Core step in predictive modeling.

12. Impact of Outliers

Description: Outliers can distort regression coefficients and predictions.

Importance: Detect and handle (remove/transform) to improve robustness.

13. Assumptions in Linear Regression

Linear regression relies on statistical assumptions:

1. **Linearity:** Relationship between features and target is linear.
2. **No Multicollinearity:** Features should not be highly correlated.
 - **VIF (Variance Inflation Factor):** Measures multicollinearity.
 - i. If $VIF > 10 \rightarrow$ problematic.
 - ii. If $5 < VIF < 10 \rightarrow$ can be taken for model building.
 - iii. If $VIF < 5 \rightarrow$ the feature is surely to be taken in the model building part.
3. **Normality of Errors:** Residuals follow normal distribution.
4. **Independence of Errors:** Errors are independent across observations.
5. **Homoscedasticity:** Error variance is constant across predictions.

Importance: Violations lead to biased or inefficient estimates.

14. Polynomial Regression

Description: Extends linear regression by including polynomial terms of features.

$$y = w_0 + w_1 x_1 + w_2 x_1^2 + w_3 x_1^3 + \dots + w_n x_1^n$$

Where, w_0 : is the intercept **and** w_n : is the weight associated with the n^{th} power of the input feature.

Importance: Captures non-linear relationships while staying interpretable.

15. Occam's Razor

Description: Prefer the simplest model that explains the data well.

Importance: Prevents overfitting by avoiding unnecessary complexity.

16. Bias-Variance Tradeoff

Description:

- **Bias:** Error due to oversimplified assumptions (underfitting).
 - **Variance:** Error due to sensitivity to fluctuations in training data (overfitting).
Importance: Guides balance between simplicity and complexity.
-

17. Regularization

Description: Technique to prevent overfitting by penalizing large coefficients.

- **Lasso (L1):** Shrinks some coefficients to zero (feature selection).
 - **Ridge (L2):** Shrinks coefficients but keeps all features.
 - **Elastic Net:** Combines both.
Importance: Improves generalization and stability of regression models.
-

18. Adjusted R^2 score

Description: a modified version of R-squared score, that increases if a new predictor improves the model by more than the expected amount.

$$\text{Adjusted } R^2 \text{ score} = 1 - \frac{(1-R^2)(N-1)}{(N-d-1)}$$

Where,

R^2 is the original R-squared score **and**

N is the total number of data points **and**

d is the number of independent variables

Importance: Provides a more reliable measure of multiple linear regression problems.